

B-dns-backup

```
• student@remote-desktop-bscne-v2:~/d522/d522-python-for-it-automation$ date 6& python3 backup_dns_config.py 6& ls -R DNS-Backup 6& head -5 DNS-Backup/Server-1/record-config.txt
Fri Aug 14 20:52:22 UTC 2026
10.10.10.10 - backup saved to /home/student/d522/d522-python-for-it-automation/DNS-Backup/Server-1/record-config.txt
10.10.10.20 - backup saved to /home/student/d522/d522-python-for-it-automation/DNS-Backup/Server-2/record-config.txt
DNS-Backup:
Server-1 Server-2

DNS-Backup/Server-1:
record-config.txt

DNS-Backup/Server-2:
record-config.txt
DNS Server: 10.10.10.10

$TTL      604800
@         IN      SOA     ns1.d522.wgu.internal. admin.d522.wgu.internal. (
2                ; Serial
```

C1-devicelist

```
• student@remote-desktop-bscne-v2:~/d522/d522-python-for-it-automation$ date && cat network_devices.csv && python3 read_devices.py
Fri Aug 14 21:59:59 UTC 2026
Device ID,Device Name,Device Address,Subnet Mask,Location,Access Port,OS,Username>Password
1001,API,10.10.10.200,255.255.255.0,Services ,5008,Ubuntu,ubuntu,ubuntu
1002,DB,10.10.10.210,255.255.255.0,Services ,5011,Ubuntu,ubuntu,ubuntu
1003,DNS1,10.10.10.10,255.255.255.0,Services ,5013,Ubuntu,ubuntu,ubuntu
1004,DNS2,10.10.10.20,255.255.255.0,Services ,5017,Ubuntu,ubuntu,ubuntu
1005,PC1,DHCP,255.255.255.0,Site1,5032,Ubuntu,ubuntu,ubuntu
1006,PC2,DHCP,255.255.255.0,Site1,5029,Ubuntu,ubuntu,ubuntu
1007,PC3,DHCP,255.255.255.0,Site1,5031,Ubuntu,ubuntu,ubuntu
1008,PC4,DHCP,255.255.255.0,Site2,5012,Ubuntu,ubuntu,ubuntu
1009,ROUTER1,10.10.10.1,255.255.255.0,Main,5021,VyOS,vyos,vyos
1010,SMTP,10.10.10.100,255.255.255.0,Services ,5019,Ubuntu,ubuntu,ubuntu
1011,SVR1,192.168.20.210,255.255.255.0,Site2,5022,Ubuntu,ubuntu,ubuntu
1012,SVR2,192.168.30.210,255.255.255.0,Site2,5014,Ubuntu,ubuntu,ubuntu
1013,SW1,None,255.255.255.0,Main,5023,OpenvSwitch,none,none
1014,SW2,None,255.255.255.0,Site1,5025,OpenvSwitch,none,none
1015,SW3,None,255.255.255.0,Site2,5027,OpenvSwitch,none,none
1016,SW4,None,255.255.255.0,Services ,5015,OpenvSwitch,none,none
API - 10.10.10.200
DB - 10.10.10.210
DNS1 - 10.10.10.10
DNS2 - 10.10.10.20
PC1 - DHCP
PC2 - DHCP
PC3 - DHCP
PC4 - DHCP
ROUTER1 - 10.10.10.1
SMTP - 10.10.10.100
SVR1 - 192.168.20.210
SVR2 - 192.168.30.210
SW1 - None
SW2 - None
SW3 - None
SW4 - None
```

C2-unavailable-email

MailHog

Not Secure http://mail.d522.wgu.internal:8025/#

It looks like you haven't started Firefox in a while. Do you want to clean it up for a fresh, like-new experience? And by the way, welcome back! Refresh Firefox...

MailHog

Search

GitHub

Connected

Inbox (1)

Delete all messages

Jim

Jim is a chaos monkey.
Find out more at GitHub.

Enable Jim

From: monitor@d522.wgu.internal

Subject: **Network Device Unavailable: SVR1 (192.168.20.210)**

To: stakeholders@d522.wgu.internal

Show headers

Plain text

Source

Dear Network Administrator,

This is an automated notification that the following network device is currently unavailable:

Device Name: SVR1
IP Address: 192.168.20.210
Last Checked: 2026-08-14 22:25:24.289955

Please investigate this issue at your earliest convenience.

Best regards,
Network Monitoring System

C3-created-ticket

```
student@remote-desktop-bscne-v2:~/ds22-python-for-it-automation$ date && python3 monitor_network.py
Fri Aug 14 22:23:32 UTC 2026
Monitoring started. Ctrl+C to stop.
API 10.10.10.200 - DNS OK 10.10.10.10
DB 10.10.10.210 - DNS OK 10.10.10.10
DNS1 10.10.10.10 - DNS OK 10.10.10.10
DNS2 10.10.10.20 - DNS OK 10.10.10.10
SVR1 192.168.20.210 - DNS OK 10.10.10.10
SVR2 192.168.30.210 - DNS OK 10.10.10.10
API 10.10.10.200 - DNS OK 10.10.10.10
DB 10.10.10.210 - DNS OK 10.10.10.10
DNS1 10.10.10.10 - DNS OK 10.10.10.10
DNS2 10.10.10.20 - DNS OK 10.10.10.10
SVR1 192.168.20.210 - DNS OK 10.10.10.10
SVR2 192.168.30.210 - DNS OK 10.10.10.10
API 10.10.10.200 - DNS OK 10.10.10.10
DB 10.10.10.210 - DNS OK 10.10.10.10
DNS1 10.10.10.10 - DNS OK 10.10.10.10
DNS2 10.10.10.20 - DNS OK 10.10.10.10
SVR1 192.168.20.210 - DNS OK 10.10.10.10
SVR2 192.168.30.210 - DNS OK 10.10.10.10
API 10.10.10.200 - DNS OK 10.10.10.10
DB 10.10.10.210 - DNS OK 10.10.10.10
DNS1 10.10.10.10 - DNS OK 10.10.10.10
DNS2 10.10.10.20 - DNS OK 10.10.10.10
UNAVAILABLE SVR1 192.168.20.210 - email sent
TICKET CREATED 201 {"assigned_to":"network-team","created_at":"2026-08-14T22:25:26Z","description":"SVR1 (192.168.20.210) did not respond to ICMP ping and is unavailable.","id":10,
"priority":"high","requester_email":"monitor@ds22.wgu.internal","status":"open","title":"Device unavailable: SVR1","updated_at":"2026-08-14T22:25:26Z"}
```

C4-dns-alert-email

MailHog

Search

GitHub

Connected

Inbox (2)

Delete all messages

Jim

Jim is a chaos monkey.
Find out more at [GitHub](#).

Enable Jim

From: monitor@d522.wgu.internal

Subject: **DNS Configuration Alert: SVR2 (192.168.30.210)**

To: stakeholders@d522.wgu.internal

Plain text Source

Dear Network Administrator,

This is an automated alert that the DNS configuration for the following device has been altered from the expected configuration.

Device Name: SVR2
IP Address: 192.168.30.210
Detected DNS Setting: 203.0.113.10
Expected DNS Setting: 10.10.10.10
Time Detected: 2026-08-14 22:38:53.716047

The system will attempt to automatically correct this configuration.

Best regards,
Network Monitoring System

C4-altered-dns

```
DNS ALTERED SVR2 detected 203.0.113.10 - email sent  
TICKET CREATED Z01 {"assigned_to":"network-team","created_at":"2026-08-14T22:38:55Z","description":"SVR2 (192.168.39.210) was found using DNS 203.0.113.10 instead of an approved internal DNS server.", "id":11, "priority":"high", "requester_email":"monitor@d522.wgu.internal", "status":"open", "title":"Unauthorized DNS change on SVR2", "updated_at":"2026-08-14T22:38:55Z"}
```

C4-corrected-dns and ticket-updated

```
DNS CORRECTED SVR2 192.168.30.210 - nameserver 10.10.10.10  
TICKET UPDATED 11 200 {"assigned_to":"network-team","created_at":"2026-08-14T22:38:55Z","description":"SVR2 DNS was automatically corrected to 10.10.10.10 by the monitoring system.", "id":11,"priority":"high","requester_email":"monitor@522.wgu.internal","status":"resolved","title":"Unauthorized DNS change on SVR2","updated_at":"2026-08-14T22:39:16Z"}
```